Exercise 5

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2026-03-15

```r
# Parameters
n <- 48
phi <- 0.5
theta <- 0.45

# Components of the asymptotic variance formula
num_phi <- (1 - phi^2) * (1 - phi * theta)
num_theta <- (1 - theta^2) * (1 - phi * theta)
denom <- (phi - theta)^2

# 1. Evaluate Variances
var_phi <- (1/n) * (num_phi / denom)
var_theta <- (1/n) * (num_theta / denom)

# 2. Evaluate Correlation
# Correlation is independent of n in the asymptotic formula
cor_phi_theta <- -(sqrt(1 - phi^2) * sqrt(1 - theta^2)) / (1 - phi * theta)

# Display Results
cat("Variance of phi-hat: ", var_phi, "\n")
```

```
## Variance of phi-hat:  4.84375
```

```r
cat("Variance of theta-hat:", var_theta, "\n")
```

```
## Variance of theta-hat: 5.150521
```

```r
cat("Correlation:          ", cor_phi_theta, "\n")
```

```
## Correlation:          -0.9979167
```

The variances are massive > 4 relative to the parameter values 0.5 and 0.45. This occurs because the denominator $(\phi - \theta)^2$ is extremely small 0.0025. As ϕ approaches θ, the variance tends toward infinity. The correlation is nearly -1. This indicates that the AR and MA components are nearly canceling each other out (the roots are too close). Mathematically, the model is almost indistinguishable from white noise. A sample of $n = 48$ is insufficient to resolve such similar AR and MA components.

```r
n <- 120

# Components of the asymptotic variance formula
num_phi <- (1 - phi^2) * (1 - phi * theta)
num_theta <- (1 - theta^2) * (1 - phi * theta)
denom <- (phi - theta)^2

# 1. Evaluate Variances
var_phi <- (1/n) * (num_phi / denom)
var_theta <- (1/n) * (num_theta / denom)

# 2. Evaluate Correlation
# Correlation is independent of n in the asymptotic formula
cor_phi_theta <- -(sqrt(1 - phi^2) * sqrt(1 - theta^2)) / (1 - phi * theta)

# Display Results
cat("Variance of phi-hat: ", var_phi, "\n")
```

```
## Variance of phi-hat:  1.9375
```

```r
cat("Variance of theta-hat:", var_theta, "\n")
```

```
## Variance of theta-hat: 2.060208
```

```r
cat("Correlation:          ", cor_phi_theta, "\n")
```

```
## Correlation:           -0.9979167
```

Even though the variance is twice less than the first experiment, compared to the parameters, we still face the same issue.